

Abstract

The persistent accumulation of selective serotonin reuptake inhibitors (SSRIs), particularly fluoxetine, poses an uncharacterized threat to non-target aquatic vegetation. This study evaluates the mechanisms of fluoxetine toxicity and exogenous melatonin mitigation using the model fern *Ceratopteris richardii*. Chronic exposure to 4.0 μM fluoxetine induced severe physiological stress, suppressing spore germination to 17.20% ($p < 0.0001$) and imposing growth arrest on cell division, rhizoid elongation, and prothallus expansion. To elucidate the underlying molecular targets, virtual molecular docking of candidate plant proteins was performed in AutoDock Vina and integrated with public RNA-seq expression profiles. Docking revealed that fluoxetine strongly binds to the ABCB1.1 multidrug transporter ($\Delta G = -8.3$ kcal/mol) concentrated in the gametophyte notch meristem, disrupting polar auxin transport required for cell division and organogenesis. Concurrently, fluoxetine targets native melatonin biosynthesis by binding to CrSNAT ($\Delta G = -7.5$ kcal/mol) and binds frontline ROS scavengers such as CrAPX1 ($\Delta G = -8.4$ kcal/mol). Co-treatment with 10.0 μM melatonin significantly rescued spore germination to 66.20% ($p < 0.0001$) and enabled post-fertilization sporophyte initiation by Day 40 despite upstream gametophytic growth restrictions. This stage-dependent rescue is governed by transcriptomic dynamics: during embryonic development, expression of the master auxin receptor CrTIR1 surges. Because melatonin exhibits a superior binding affinity for CrTIR1 ($\Delta G = -8.5$ kcal/mol) over fluoxetine ($\Delta G = -8.1$ kcal/mol), it outcompetes the contaminant to restore essential auxin signaling cascades. This study establishes a multi-omic framework for remediating pharmaceutical pollution in aquatic networks.